Natural Language Processing

CS4063

ASSIGNMENT 2

National University
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Overview

This report documents the implementation and evaluation of a full neural NLP pipeline built from scratch in PyTorch on a BBC Urdu corpus of 300 articles. The pipeline spans three parts: word embedding learning (TF-IDF, PPMI, and Skip-gram Word2Vec), sequence labelling using a BiLSTM tagger for POS and NER, and topic classification using a custom Transformer encoder. No pre-trained models, Gensim, or HuggingFace libraries were used. All implementation strictly follows the assignment constraints.

Part 1 Results: Word Embeddings

1.1 TF-IDF Weighting

A term-document matrix was constructed from cleaned.txt with vocabulary capped at 10,000 most frequent tokens; all others map to <UNK>. TF-IDF weights were computed using the standard formula and the resulting matrix (shape 10,001 × 300) was saved as tfidf_matrix.npy. The top-10 most discriminative words per topic based on mean TF-IDF scores are shown below (a representative selection):

```
Top-10 most discriminative words per topic (mean TF-IDF):

Politics:
  1. حکومت
  2. معدانی
  3. میئر
  4. پختونخوا
  5. چراوہے
  6. بکریاں
  7. خیبر
  8. وفاقی
  9. چرانے
 10. سعودی

Sports:
  1. کرکٹ
  2. میچ
  3. رنز
  4. ٹیم
  5. بنگلہ
  6. دیش
  7. کپ
  8. انڈیا
  9. پی
 10. ورلڈ

Economy:
  1. کنٹکٹر
  2. چپس
  3. سیمی
  4. کرپٹو
  5. فونز
  6. اونٹوں
  7. بٹ
  8. بینک
  9. سعودی
 10. اونٹ

International:
  1. ایران
  2. جاتیے
  3. ویڈیو
  4. مظاہرین
  5. پروڈیوسر
  6. امریکہ
  7. برف
  8. سیربین
  9. چالان
 10. چین

Health_Society:
  1. تھراپی
  2. صحت
  3. مریضوں
  4. میوزک
  5. بچوں
  6. شادی
  7. سلگتی
  8. آفتیں
  9. کہتی
 10. ایشیا
```

1.2 PPMI Co-occurrence Matrix

A word-word co-occurrence matrix was built from cleaned.txt using a symmetric context window of size k = 5, restricted to the top 2,000 vocabulary tokens. Positive PMI was applied and the resulting 2,000 × 2,000 matrix saved as ppmi_matrix.npy. A t-SNE visualisation of the 200 most frequent tokens was produced, colour-coded by topic category with a legend. The top-5 nearest neighbours for 10 query words by cosine similarity are shown below:

```
Top-5 nearest neighbours in PPMI space:

  پاکستان: ['انڈیا', 'کرکٹ', 'ٹیم', 'میچ', 'کپ']
  کرکٹ: ['بورڈ', 'ٹیم', 'پاکستان', 'میچ', 'کپ']
  حکومت: ['کے', 'وزیر', 'سعودی', 'کو', 'جانب']
  معیشت: ['تیل', 'ملکی', 'کاری', 'انحصار', 'صنعت']
  فوج: ['ایرانی', 'فوجی', 'ایران', 'امریکی', 'امریکہ']
  صحت: ['ذہنی', 'مند', 'بے', 'ایچ', 'یا']
  تعلیم: ['یافتہ', 'یونیورسٹی', 'سکول', 'مابر', 'کے']
  بینک: ['جمع', 'سامان', 'وہاں', 'تک', 'اندر']
  عدالت: ['کورٹ', 'ہائی', 'نے', 'کو', 'گیا']
  سیاست: ['سیاسی', 'سے', 'نہیں', 'تو', '۔']
```

2.1 Skip-gram Word2Vec

A Skip-gram model was trained with separate centre (V) and context (U) embedding matrices of dimension |V| × d. Negative sampling used a noise distribution P(w) ∝ f(w)^0.75 with K = 10 samples per positive pair. The BCE loss was optimised with Adam (lr = 0.001) over 5 epochs and batch size 512. All required hyperparameters were met: d = 100, k = 5, K = 10. Three training conditions were run:

- C3: cleaned.txt, d=100: 3,710,330 training pairs, loss decreasing from 4.27 → 2.87 over 5 epochs
- C2: raw.txt, d=100: 3,569,930 pairs, loss from 4.22 → 2.76
- C4: cleaned.txt, d=200: same pairs as C3, loss from 4.03 → 2.82

Averaged embeddings ½(V+U) from C3 were saved as embeddings_w2v.npy. Loss curves were plotted for all conditions.

2.2 Evaluation

Nearest Neighbours

Top-10 nearest neighbours were computed for all 8 required query words using cosine similarity on C3 embeddings. Notable results: عدالت (court) retrieved مقدمے (cases), جسٹس ,کورٹ with high

similarity (0.83+), demonstrating strong legal domain clustering. فوج (army) retrieved مسلح ,ایرانی, تہران, indicating geopolitical context. آبادی (population) was not present in the vocabulary due to low frequency in the corpus.

Analogy Tests

Ten analogy tests of the form v(b) − v(a) + v(c) were constructed. The model scored 5/10 correct, meeting the minimum threshold. Successful analogies included politics→legal (وزیر:حکومت :: جج:عدالت), sports→military (فوج:مسلح :: میچ:کرکٹ), and education→health (مدرسے:تعلیم :: صحت:ذہنی). The five failures were concentrated in health/economy domains and geography→military, where limited corpus coverage prevented the model from learning reliable analogy relationships.

Four-Condition Comparison (MRR on 20 Labelled Pairs)

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Four-Condition Comparison — MRR on 20 labelled pairs

ID    Condition                              MRR

C1    PPMI baseline                        0.2282
C2    Skip-gram on raw.txt (d=100)         0.2048
C3    Skip-gram on cleaned.txt (d=100)     0.2127
C4    Skip-gram on cleaned.txt (d=200)     0.1981
```

C1 (PPMI) achieved the highest MRR, which is expected on a small 300-article corpus where count-based methods are more sample-efficient. C3 outperforms C2, confirming that text cleaning is the most impactful preprocessing step. C4 underperforms C3, showing that doubling dimensionality without sufficient data introduces sparsity rather than richer representations. On a corpus of 10,000+ articles, C4 would be expected to outperform C3.

Part 2 Results: Sequence Labeling

3. Dataset Preparation

500 sentences were selected from cleaned.txt using topic-stratified sampling. Three topics exceeded the 100-sentence minimum: Sports (171), Politics (144), and International (140). Economy (12) and Health_Society (33) had limited sentence counts due to their smaller article base. The 70/15/15 split yielded Train: 348 | Val: 72 | Test: 80 sentences.

A rule-based POS tagger was built using a hand-crafted lexicon of 683 entries with suffix-based morphological fallback rules. NER annotation used the BIO scheme backed by a gazetteer of 55 Pakistani persons, 75 locations, and 40 organisations, all meeting the assignment minimums.

```
Train: 348 | Val: 72 | Test: 80

POS tag distribution (train):
  ADJ   : 521
  ADV   : 192
  CONJ  : 601
  DET   : 71
  NOUN  : 3609
  NUM   : 149
  POST  : 1311
  PRON  : 944
  PUNC  : 9
  VERB  : 1810

NER tag distribution (train):
  B-LOC   : 134
  B-MISC  : 3
  B-ORG   : 43
  B-PER   : 3
  I-LOC   : 11
  I-ORG   : 69
  I-PER   : 3
  O       : 8951
```

4 & 5.1 POS Tagging Results

The BiLSTM tagger was trained in both frozen and fine-tuned embedding modes using C3 Word2Vec initialisations. Fine-tuned embeddings significantly outperformed frozen ones across all metrics. The model was trained with Adam (lr=1e-3, weight decay=1e-4), dropout p=0.5, and early stopping on validation F1 (patience=5).

```
POS — Frozen vs Fine-Tuned Embedding Comparison
Mode              Accuracy   Macro-F1

Frozen              0.8104     0.5044
Fine-Tuned          0.9331     0.7512
```

The test set confusion matrix revealed three major error patterns. ADJ→NOUN was the most common (43 errors), as many Urdu adjectives like سعودی appear without their noun head in isolation. VERB→NOUN (32 errors) occurred for ambiguous forms such as اعلان which can serve as either. NOUN→ADJ (24 errors) reflects the same morphological ambiguity.

5.2 NER Results

Entity-level evaluation using conlleval-style counting was performed for both CRF and softmax output conditions. Both configurations scored 0.0 across all entity types, indicating the model defaulted to predicting only O tags. This is a direct consequence of extreme label sparsity in the

training data, only 263 named entity tokens across 9,214 total training tokens (2.9%), which is insufficient signal for the BiLSTM to learn reliable B/I boundaries.

5.3 Ablation Study

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Ablation Study Results — Macro F1

  A1 (Uni-LSTM)                : 0.5387
  A2 (No Dropout)              : 0.7602
  A3 (Random Emb)              : 0.6234
  A4 (NER No CRF)              : 0.1098
  Baseline (BiLSTM+CRF)        : 0.7512
```

A1 (Unidirectional LSTM) scored test macro-F1 of 0.5387 vs the baseline of 0.7512, a drop of 0.1125, confirming that backward context is important for Urdu POS tagging where verbs and postpositions appear sentence-finally. A2 (No Dropout) surprisingly scored 0.7602, slightly above the baseline. This is counter-intuitive but explainable: without dropout the model memorised the small 348-sentence training set more aggressively, and the validation F1 kept climbing to 0.7854 by epoch 20 without early stopping triggering, meaning it overfits but the test set happened to benefit. A3 (Random Embeddings) scored 0.6234, below the pre-trained baseline, confirming that Word2Vec initialisation from Part 1 provides a meaningful head start. A4 (NER without CRF) scored token-level macro-F1 of 0.1098 (matching O-class dominance), and entity-level F1 of 0.0000, identical to the CRF variant. The root cause is label sparsity, not decoding strategy.

Part 3 Results: Transformer Encoder for Topic Classification

6. Dataset Preparation

All 300 articles were labelled into 5 topic categories using keyword-based scoring against the assignment's indicative keyword lists. Each article was represented as a token-ID sequence padded or truncated to 256 tokens. The 70/15/15 stratified split produced Train: 207 | Val: 43 | Test: 50.

```
Train: 207 | Val: 43 | Test: 50

Class distribution (train):
  Politics: 39
  Sports: 63
  Economy: 10
  International: 65
  Health_Society: 30
```

```
Class distribution (val):
  Politics: 8
  Sports: 13
  Economy: 2
  International: 14
  Health_Society: 6

Class distribution (test):
  Politics: 9
  Sports: 15
  Economy: 3
  International: 15
  Health_Society: 8
```

7. Transformer Architecture

All Transformer components were implemented as independent modules without using nn.Transformer, nn.MultiheadAttention, or nn.TransformerEncoder. The architecture consists of:

- Scaled dot-product attention: accepts optional padding mask, returns output and attention weights
- Multi-head self-attention: h=4 heads, d_model=128, d_k=d_v=32, separate projection matrices per head with shared output projection W_O
- Position-wise FFN: two linear layers with ReLU, inner dimension d_ff=512, dropout between layers
- Sinusoidal positional encoding: stored as a non-learned buffer via register_buffer()
- 4 stacked Pre-Layer Norm encoder blocks: $x \leftarrow x + \text{Dropout}(\text{MHSA}(\text{LN}(x)))$, $x \leftarrow x + \text{Dropout}(\text{FFN}(\text{LN}(x)))$
- Classification head: learnable [CLS] token prepended, MLP 128 → 64 → 5 applied to CLS output

Total trainable parameters: 2,080,261. Training used AdamW ($lr=5\times10^{-4}$, weight decay=0.01) with a cosine LR schedule and 50 warmup steps over 20 epochs.

8.1 Classification Results

```
Test Accuracy : 0.5400
Test Macro-F1 : 0.4015

Per-class Classification Report:
                precision    recall  f1-score   support

      Politics       0.43      0.33      0.38         9
        Sports       0.76      0.87      0.81        15
       Economy       0.00      0.00      0.00         3
 International       0.41      0.60      0.49        15
Health_Society       0.50      0.25      0.33         8

      accuracy                           0.54        50
     macro avg       0.42      0.41      0.40        50
  weighted avg       0.51      0.54      0.51        50
```

Sports achieved the highest F1 (0.81) due to its large training sample and distinctive vocabulary. Economy scored 0.00 F1, with only 10 training articles it lacked sufficient signal. Attention heatmaps from 2 heads of the final encoder layer for 3 correctly classified articles confirmed that the [CLS] token attends to domain-specific keywords: cricket and team tokens for sports articles, government and minister tokens for politics articles.

8.2 BiLSTM vs. Transformer Comparison

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BiLSTM vs. Transformer — Topic Classification Summary

Metric                        BiLSTM  Transformer

Test Accuracy                 0.5000       0.5400
Test Macro-F1                 0.4173       0.4015
Epochs trained                    20           20
Best Val Acc                  0.6512       0.4884
```

1. The Transformer achieved higher test accuracy (0.5400 vs 0.5000, +4 points) but lower macro-F1 (0.4015 vs 0.4173), indicating it performs better on majority classes while the BiLSTM handles minority classes more evenly.

2. The BiLSTM converged faster in terms of best validation accuracy, peaking at 0.6512 at epoch 9 before degrading from overfitting. The Transformer's best val accuracy of 0.4884 was reached around epoch 10 and plateaued, suggesting it never fully generalised on this small corpus.

3. The BiLSTM was significantly faster per epoch. LSTM recurrence runs in optimised CUDA kernels with low parameter count. The Transformer computes $O(T^2)$ dot products per attention layer across T=256 tokens with 4 stacked blocks, making each forward pass considerably heavier.

4. Attention heatmaps showed the [CLS] token concentrating on domain-relevant keywords — sports articles attended to کرکٹ and ٹیم, politics articles to حکومت and وزیر. This confirms the model routes meaningful signals into the classification head through self-attention.

5. Given only 200-300 articles, the BiLSTM is the more appropriate architecture. Its sequential inductive bias and pre-trained Word2Vec initialisation give it a strong starting point. Transformers are data-hungry — self-attention has no positional or sequential inductive bias and must learn all structure from data, which 300 articles cannot adequately provide.

Conclusion

The TF-IDF and PPMI representations captured meaningful topic-level structure, and Skip-gram Word2Vec embeddings demonstrated semantic clustering across legal, military, and sports domains. The BiLSTM POS tagger achieved 93.3% accuracy with fine-tuned embeddings, confirming the value of pre-training and embedding adaptation. NER performance was limited by extreme label sparsity in the gazetteer-annotated training data. The custom Transformer encoder achieved competitive classification accuracy and its attention maps revealed interpretable topic focused patterns, though the BiLSTM remained the more sample-efficient architecture for this corpus size.